

Introduction to Cyber crime

Cybercrime definitions

- A crime conducted in which a computer was directly and significantly instrumental is known as “**Computer Crime**”.
- Computer crime as also other various definitions :
 - Any threats to the computer itself, such as theft of hardware or software and demands for ransom.
 - Any financial dishonesty that takes place in a computer environment.

Cyber Crime definitions...

- A crime committed using a computer and the Internet to steal person's identity or sell illegal or smuggled goods or disturb any operations with malicious program is known as **“Cyber Crime”**.
- Another definition is :
 - Any illegal activity done through the internet.
 - Any criminal activities done using cyberspace and WWW.

The first Cyber Crime

- **The first recorded cyber crime took place in 1820.**
- *In 1820, Joseph-Marie Jacquard, a textile manufacturer in France, produced the loom. This device allowed the repetition of a series of steps in the weaving of special fabrics. This resulted in a fear amongst Jacquard's employees that their traditional employment and livelihood were being threatened. They committed acts of sabotage to discourage Jacquard from further use of the new technology. This is the first recorded cyber crime!*

Cyberspace

- Cyberspace is a world-wide network of computer networks that uses the TCP/IP for communication to facilitate transmission and exchange of data.
- A “nebulous place” where humans interact over computer networks.
- Internet and other computer networks.

Cyberspace

- Cyberspace is a place where you can chat, explore, research and play (INTERNET).

Cyber squatting

- This term is derived from “squatting” which is the act of occupying an vacant/unoccupied space that the squatter does not own or rent.
- Cybersquatting is the act of registering a popular Internet address--usually a company name--with the intent of selling it to its rightful owner.

Cyberpunk

- The word “cyber” and “punk” are two different words which means “disorder via machine”.
- The word cyberpunk was coined by writer Bruce Bethke, who wrote a story with that title in 1982. He derived the term from the words cybernetics, the science of replacing human functions with computerized ones, and punk, the harsh music that developed in the youth culture during the 1970s and '80s.

- The movies based on cyberpunk are :
 - Terminator I, II and III
 - Until the end of the world
 - Mad MAX I, II and III
 - The Matrix (series)
 - The X-Files
 - Solaris

Cyber warfare

- Defⁿ: Information warriors unleashing vicious attacks against an unsuspecting opponent's computer networks, wreaking havoc and paralyzing nations.
- Cyber warfare refers to politically motivated hacking.
- Attack on information infrastructure.
- information infrastructure refers to information resources, including communication systems that support an industry, institution or population.
- Example :
 - Pentagon planning.

Cyberterrorism

- Cyberterrorism is the premeditated , politically motivated attack against information, computer systems, computer programs and data which result in violence against noncombatant targets by sub national groups or clandestine agents.
- Cyber terrorism is “any person, group or organization who with terrorist intent, utilizes, accesses or aids in accessing a computer or computer network or electronic system or electronic device by any available means and there by knowingly engages in a terrorist act.
- Use of information technology and means by terrorist groups and agents

Diff. between Cyber Crime and terrestrial crime

- It differs in four ways.

1. How to commit them is easier to learn
2. They require few resources relative to the potential damage caused.
3. They are often not clearly illegal.
4. They can be committed without being physically present.

Cybercrime and Information Security

- Lack of information security gives rise to cyber crime.
- Cyber security means protecting information, equipment, devices, computer, computer resource, communication device and information stored in all these from unauthorized access, use, disclosure, disruption, modification or destruction.

Who are Cybercriminals

- Cybercrime involves such activities like
 - Child pornography
 - Credit card fraud
 - Cyberstalking (irritation)
 - Defaming another online
 - Gaining unauthorized access to computer system
 - Overriding encryption to make illegal copies
 - Software piracy
 - Stealing another's identity to perform criminal act.

Types of Cybercriminals

- Type 1 : Hungry for recognition
- Type 2 : Not interested in recognition
- Type 3 : The insiders

Type 1: Hungry for recognition

- Hobby hackers
- IT professionals
- Politically motivated hackers
- Terrorist organizations.

Type 2: Not interested in recognition

- Psychological spoiled
- Financially motivated hackers
- State-sponsored hacking
- Organized criminals

Type 3 : The insiders

- Disgruntled or former employees seeking revenge
- Competing companies using employees to gain economic advantage through damage / theft.

Classifications of Cybercrimes

- Cybercrime against individual
- Cybercrime against property
- Cybercrime against organization
- Cybercrime against society
- Crimes start from Usenet newsgroup

Cybercrime against individual

- Email spoofing
- Phishing
- Spamming
- Cyberdefamation
- Cyberstalking and harassment
- Pornographic offense
- Password sniffing

Cybercrime against property

- Credit card frauds
- Intellectual Property Crime
- Internet time theft

Cybercrime against organization

- Unauthorized accessing of computer
- Password sniffing
- Denial-of-service attacks
- Email bombing

- Salami attack
- Logic bomb
- Trojan Horse
- Data diddling

- Crimes starts from Usenet newsgroup
- Industrial spying
- Computer network disturbance
- Software piracy

Cybercrime against society

- Forgery
- Cyberterrorism
- Web jacking

E-Mail Spoofing

- A spoofed E-mail is one that appears to originate from one source but actually has been sent from another source.

Example

- A branch of global trust bank experienced a customer spreads out the rumor that bank is not doing well.

Spamming

- People who create electronic spam are called **“Spammers”**.
- Spam is the abuse of e-messaging systems to send unsolicited (unwanted) bulk messages.
- Spamming is difficult to control.

- The another definition of spamming is in the context of “search engine spamming”.
- To avoid spamming, following web publishing techniques should be avoided.

- Repeating keywords
- Use of keywords that do not relate to the content on the site
- Redirection
- Duplication of pages with different URLs
- Hidden links

Cyberdefamation

- The Indian Penal Code says about defamation is
 - ***“Whoever by words either spoken or intended to be read, or by signs or by visible representations, makes or publishes any imputation concerning any person intending to harm or knowing or having reason to believe that such imputation will harm, the reputation of such person, is said, except in the cases hereinafter expected, to defame that person.”***

- When above happens in electronic form, its known as Cyberdefamation
- ***Libel*** is written defamation and ***Slander*** is oral defamation.

Internet Time Theft

- Such theft occurs when an unauthorized person uses the Internet hours paid by another person.
- Basically, internet time theft comes under hacking.

Salami Attack

- These attacks are used for committing financial crimes.
- The main idea here is to make the alteration so insignificant that in a single case it would go completely unnoticed.

- For ex. A bank employee inserts a program, into bank's servers, that deducts a small amount of money from the account of every customer. No account holder will notice this unauthorized debit, but the bank employee will make sizable amount.

Data diddling

- A data diddling attack involves altering raw data just before it is processed by a computer and then changing it back after the processing is completed.

Forgery

- Fake currency notes, postage and revenue stamps, marksheets can be forged using sophisticated computers, printers and scanners.

Web Jacking

- Web jacking occurs when someone forcefully takes control of a website.
- First stage of this crime involves “password sniffing”.

Newsgroup Spam

- This is one form of spamming.
- The first widely recognized Usenet spam titled “***Global Alert for All : Jesus is Coming Soon***” was posted on 18th January, 1994 by ***Clarence L. Thomas IV*** a sysadmin at Andrews University.

Industrial Spying

- The internet and privately networked systems provide new and better opportunities for spying.
- “Spies” can get information about product finances, research and development and marketing strategies.
- This activity is known as **“industrial spying”**.

- One of the interesting case is about The famous Israeli Trojan story, where a software engineer in London created a Trojan Horse program specifically designed to extract critical data gathered from machines infected by his program.
- He had made a business out of selling his Trojan Horse program to companies in Israel, which would use it for industrial spying by planting it into competitor's network.

Hacking

- Purpose of hacking are many, the main ones are as follows :
 - Greed (Hunger)
 - Power
 - Publicity
 - Revenge
 - Adventure
 - Desire to access forbidden information
 - Destructive mindset

- Hackers write or use ready-made computer programs to attack the target computer.
- Government websites are hot on hacker's target lists and attacks on government websites receive wide press coverage.

- In December 2009, NASA's site was hacked by SQL Injection.

Online Frauds

- This comes under spoofing. The purpose of spoofing is to make the user enter personal information which is then used to access business and bank accounts.
- This kind of online fraud is common in banking and financial sector.
- It is strongly recommended not to input any sensitive information that might help criminals to gain personal information.

- Story of Nadya Suleman and her eight babies. Hacked by MOD (Mother of Disappointment).
- CIA (Central Intelligence Agency), the US website was hacked.
- Dept. of Justice site defaced.
- Pentagon, the US site defaced.
- Twitter site hacked.

Pornographic Offenses

- The internet is being highly used by its abusers to reach and abuse children sexually, worldwide.
- “Pedophile” are people who are sexually attracted to children . They are physically and psychologically forcing minors to engage in sexual activities.

How they operate?

- Pedophiles use a false identity to trap the children/teenagers.
- They seek teens in the kids' areas.
- They be friend of them.
- Then they get email address of the child and start making contacts on email too. These emails contains sexually explicit language.

- They start sending pornographic images/text to the victim to feed into his/her mind that “This is normal and everybody does it”.
- At the end of it, the pedophiles set up a meeting with the child out of the house and then use them as a sex object.

Software Piracy

- This the “The Biggest” challenge area.
- Software piracy is ***“theft of software through the illegal copying of genuine programs or the fake program and distribution of products intended to pass for the original”***.

Disadvantage of piracy

- The software, if pirated, may potentially contain hard-drive infection virus.
- There is no technical support in the case of software failure.
- There is no warranty protection
- There is no legal right to use the product.

- According to the fourth annual BSA (Business Software Alliance) and IDC global Software Piracy study, in Asia pacific 55% of software installed are illegal.

Computer Sabotage

- Inserting worms, viruses or logic bomb in computer is referred as computer sabotage.
- Logic bomb is event dependent program created to do something only when a certain event occurs.
- Example CIH (Chernobyl virus).

Email bombing

- It refers to sending a large number of e-mails to the victim to crash victim's email account or to make victim's mail server crash.

Usenet Newsgroup

- Usenet is a mechanism that allows sharing information in a many-to-many manner.
- Usenet mainly used for following crime :
 - Distribution/sale of pornographic material
 - Distribution/sale of pirated software
 - Distribution of hacking software
 - Sale of stolen credit card number
 - Sale of stolen data

Password Sniffing

- Password sniffers are programs that monitor and record the name and password of network users as they login.

Credit Card Fraud

- Information security requirements for credit cards have been increased recently.
- Millions of dollars lost by consumers who have credit card stolen from online database.

Identity theft

- Identity theft is a fraud involving another person's identity for an illegal purpose.

Computer Network Intrusions

- Hackers can break into computer systems from anywhere in the world and steal data, plant viruses, insert trojan horses or change user names and passwords.

Questions

1. What is CyberCrime? How do you define it?
2. How do we classify cybercrimes? Explain each one briefly.
3. What are the different types of cybercriminals? Explain each one briefly.
4. State the difference between “cybercrime” and “cyberfraud” if any.

5. Define the following terms

1. Cyberterrorism
2. Cyberpunk
3. Cyberdefamation
4. Cyberwarefare